

Ultimate Nutrition

R-ALA

Alpha lipoic acid (ALA) is known by a variety of names, including thioctic acid, 1, 2-dithiolane-3-pentanoic acid, 1, 2-dithiolane-3 valeric acid, dihydrolipoic acid, lipoic acid, and 6, 8-thioctic acid. The importance of ALA was first discovered in the 1950s, when it was recognized as a powerful antioxidant and an important cofactor in the Krebs cycle (i.e., citric acid cycle), the body's main process for converting carbohydrates into energy. Put simply, ALA converts glucose (i.e., blood sugar) into energy for the body's normal function.

ALA is a sulfurous fatty acid that would be classified as a vitamin except for the fact that it can be readily synthesized within the human body. ALA can also be obtained in our diets from both animal and plant sources. It is found in small amounts in foods such as spinach, broccoli, peas, Brussels sprouts, rice bran, red meat, organ meats (e.g., liver, kidney and heart), and yeast (particularly Brewer's yeast). Although healthy individuals are able to synthesize enough ALA to meet their needs for enzyme cofactors, the supply is generally used up in carrying out metabolic functions with very little left for antioxidant function.¹ (**Note:** Antioxidants neutralize potentially harmful chemicals called free radicals, which are waste products created when the body turns food into energy. Free radicals cause harmful chemical reactions that can damage cells in the body. For this reason, supplementation proves to be beneficial especially for those who have increased oxidative stress. It has been shown that exercise produces an imbalance between free radicals, such as reactive oxygen species and antioxidants, which typically results in increased oxidative stress. Whether the body's natural antioxidant defence system is sufficient to counteract the increase in reactive oxygen species or whether additional exogenous supplements are needed is not known, although trained athletes who received antioxidant supplements show evidence of reduced oxidative stress.² Therefore, supplementing with antioxidants provides a tremendous benefit.

ALA is a unique and powerful molecule considered to be an ideal antioxidant because it can be absorbed from our diets, it can be converted in our cells into a usable form, it possesses a variety of antioxidant actions, and it has low toxicity.³ While other well-known antioxidants work only in water (such as vitamin C-protecting the cell fluid) or work only in fatty tissues (such as vitamin E-protecting cell membranes), ALA is the only antioxidant that is both fat and water soluble. ALA can access all parts of the cell, giving it tremendous ability to trap free radicals.

Moreover, there is growing evidence that ALA has the ability to regenerate other antioxidants that are depleted after removing free radicals.⁴ Normally, once an antioxidant removes free radicals, they become depleted. Evidence suggests that ALA is able to recycle antioxidants (such as vitamin C and glutathione), regenerating and reactivating them after they have been depleted. Therefore, ALA plays an important role in the synergism of antioxidants within the body. Unlike other antioxidants which are antioxidant only in their reduced form, both the oxidized and reduced forms of lipoic acid are antioxidant.⁵ ALA can also directly scavenge reactive oxygen species.

ALA contains an asymmetric carbon, resulting in two possible optical isomers (R-ALA and S-ALA) called enantiomers or "mirror images." They are chemically identical, yet certain common elements are in different positions. Like your right and left hands, they look the same but cannot be superimposed upon one another. Only the R-isomer is created by the body⁵ and considered more bioavailable (i.e., better absorbed and utilized in the body). Research has shown that after oral

dosing with a racemix (50/50 mixture) of ALA, peak plasma concentration of R-ALA was found to be 40-50% higher than S-ALA, suggesting that R-ALA is better absorbed.⁶ Therefore, supplements containing R-ALA can be considered more bioavailable. Many commercially available ALA products do not list the form of ALA. These products are likely to have inferior benefits because of lower bioavailable ingredients compared to R-ALA by Ultimate Nutrition® which uses only the R-version of lipoic acid.

ALA has also been found to affect carbohydrate metabolism through its involvement in various metabolic mechanisms. Experiments with ALA have found increased expression and translocation of the glucose transporter proteins (GLUT 4), resulting in significantly increased muscle glucose uptake.⁷ ALA appears to engage the insulin-signalling pathway, thereby increasing glucose uptake into muscle.⁸ Based on the literature, ALA speeds the removal of glucose from the blood stream at least partly by enhancing insulin function.⁵ Since skeletal muscle tissue is the major sink in the body for glucose following a meal, agents that enhance glucose uptake by skeletal muscle are very beneficial, especially during times surrounding exercise when energy stores need to recover.

Supplementing with ALA appears to be very beneficial if used in conjunction with the proper nutrients. By increasing ALA levels in the body, you support insulin activity that helps shuttle nutrients in the body. In fact, results from one study indicated that co-ingestion of ALA with creatine plus carbohydrates enhanced the total creatine content in muscle compared to creatine plus carbohydrates and creatine alone.⁹ Significant improvements in creatine content can enhance skeletal muscle exercise capacity and performance. Therefore, we can extrapolate that by combining ALA with a pre-, intra- or post-workout supplement, you optimize your body's ability to use the nutrients and recovery faster. Future research is required to determine whether this combination will lead to measurable gains in both strength and muscularity.

ALA is produced endogenously and is consumed in our diets; however, evidence suggests that supplemental ALA provides many associated benefits if timed correctly and used in conjunction with the proper nutrients. ALA is a unique and versatile antioxidant that increases nutrient delivery, deactivates an unusually wide array of cell-damaging free radicals, regenerates endogenous antioxidants, optimizes insulin activity, and supports activity recovery. For these reasons, supplementing with ALA appears to be a very beneficial nutrient if you are an active individual participating in recreational fitness or resistance exercise.

FAQs

Q: How much R-ALA should I take a day?

A: It is recommended to consume 200 mg of R-ALA with each meal.

Q: Are there any side effects associated with R-ALA?

A: Research has shown that supplementing with R-ALA in moderate consumption is relatively safe. Side effects are generally rare, and few have been reported. Rare cases may include headache, tingling or a "pins and needles" sensation, skin rash, or muscle cramps. Very high doses have been known to cause nausea, upsets stomach, vomiting, and excess doses can lead to low blood sugar levels.

Q: Can I use R-ALA if I'm taking prescription drugs?

A: R-ALA shouldn't interfere with your prescription drugs, but it's always a good idea to consult with your doctor or health care provider before starting any supplement if you are taking any medications or prescription drugs.

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